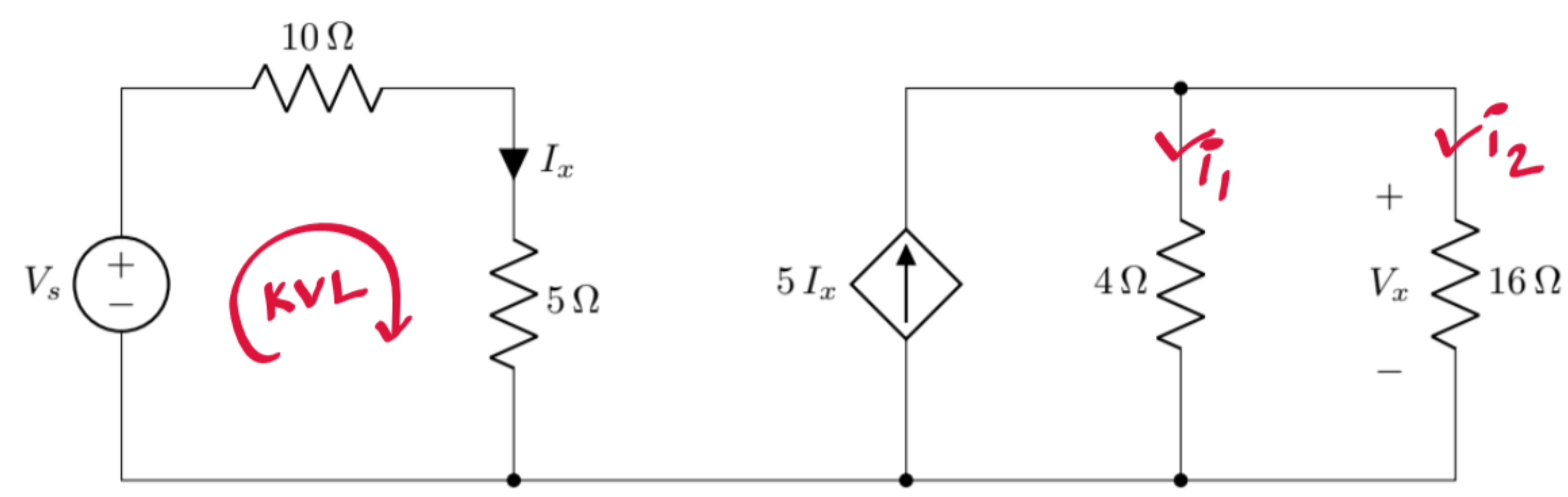
# CSE250 Summer 2025

## Set A

1)




a. $P = 16W$

Using $P = V^2/R$,

$\Rightarrow 16 = V_x^2/16$

$\boxed{\therefore V_x = 16\text{ V}}$

c. KVL in highlighted mesh,

$-V_S + 10I_x + 5I_x = 0$

$[I_x = 1A] \therefore V_S = 15 \times 1$

$\boxed{= 15\text{ V}}$

b. $5I_x$ divides between $4\Omega$ and $16\Omega$; using CDR,

$i_2 = \frac{R_{eq}}{R_{16}} \times 5I_x$

$\Rightarrow i_2 = \frac{4 \parallel 16}{16} \times 5I_x$

$= \frac{1}{5} \times 5I_x$

$= I_x$

using Ohm's Law,

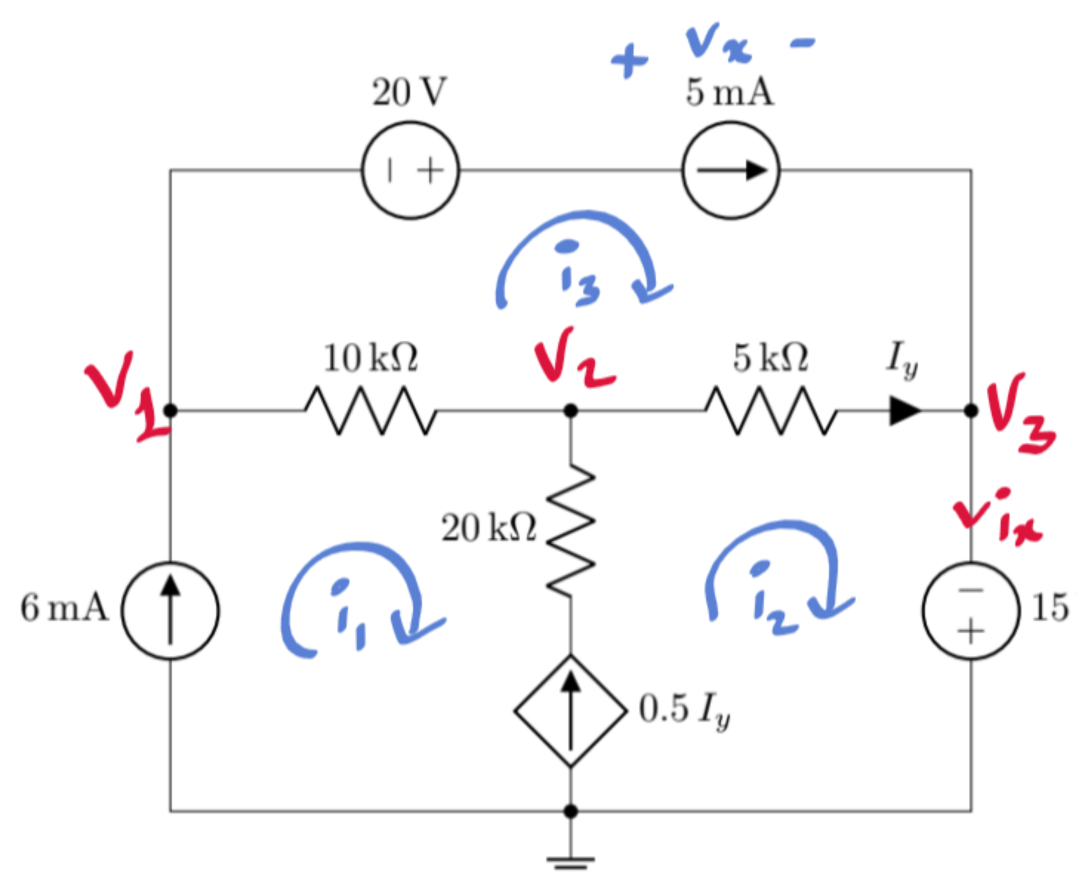
$V_x = i_2 \times 16$

$\Rightarrow 16 = i_2 \times 16$

$\therefore i_2 = 1A$

and $\boxed{I_x = 1\text{ A}}$

2)




b. see part (a).

a. Method 1 : Nodal Analysis

$V_1 : (-6) + \frac{V_1 - V_2}{10} + 5 = 0$

$\Rightarrow V_1 - V_2 = 10$ — (i)

$V_2 : \frac{V_2 - V_1}{10} + (-0.5I_y) + \frac{V_2 - V_3}{5} = 0$

$\Rightarrow -2V_1 + 10V_2 - 2V_3 = I_y$ — (ii) becomes $-V_1 + 2V_2 - V_3 = 0$

$V_3 : \quad V_3 = -15$ — (iii)

$I_y = \frac{V_2 - V_3}{5}$ — (iv)

solving,

$\boxed{V_1 = 5V,\ V_2 = -5V,\ V_3 = -15V}$

Method 2 : Mesh Analysis

(1): $i_1 = 6$ — (i)

(3): $i_3 = 5$ — (ii)

(There is no need to write an equation for mesh 2)

and KCL equations -

(2 and 3) $I_y = i_2 - i_3$ — (iii)

(1 and 2) $0.5I_y = i_2 - i_1$ — (iv) becomes $-2i_1 + i_2 + i_3 = 0$

solve (iv) -

$\therefore i_2 = 7\text{ mA}$

and $i_1 = 6\text{ mA}$

$i_3 = 5\text{ mA}$

i) $P_{5mA} = +5 \times V_x$

nodal

$= 5 \times (V_1 - V_3 + 20)$

$= 200\text{ mW}$

(absorbed)

mesh

$= +5[20 - 5(i_3 - i_2) - 10(i_3 - i_1)]$

$= 200\text{ mW}$

(absorbed)

ii) $P_{15V} = -i_x \times 15$

nodal

$= -(5 + I_y) \times 15$

$= -105\text{ mW}$

(supplied)

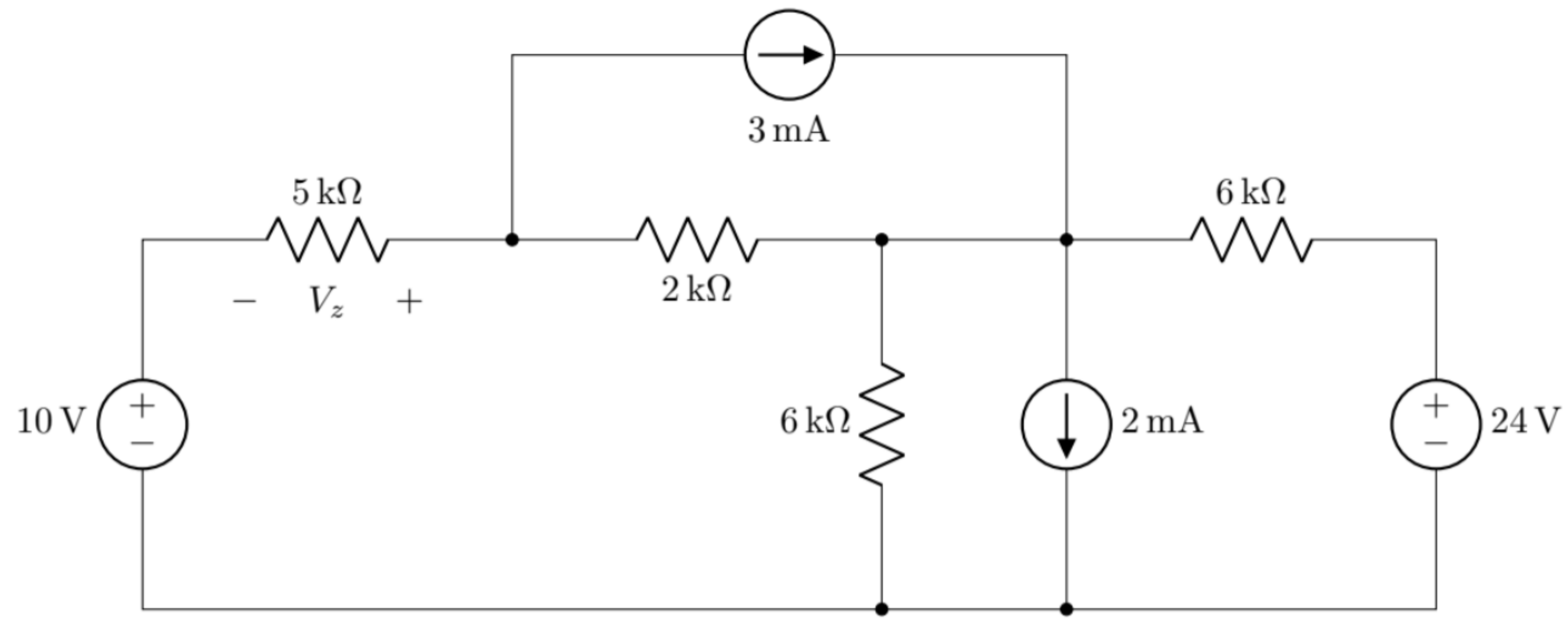
mesh

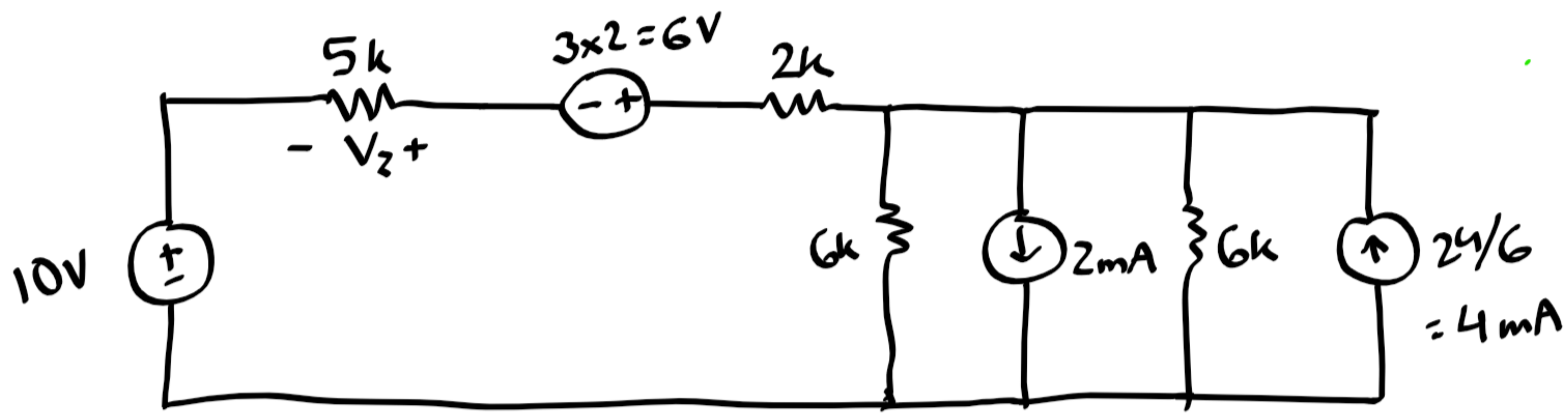
$= -i_2 \times 15$

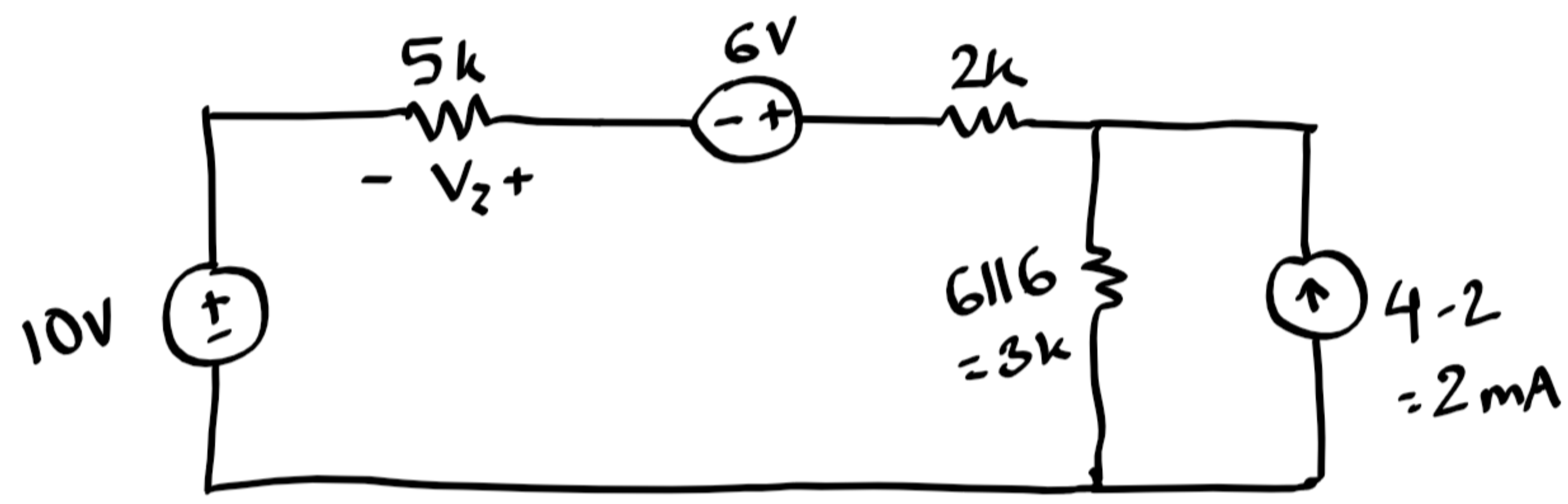
$= -105\text{ mW}$

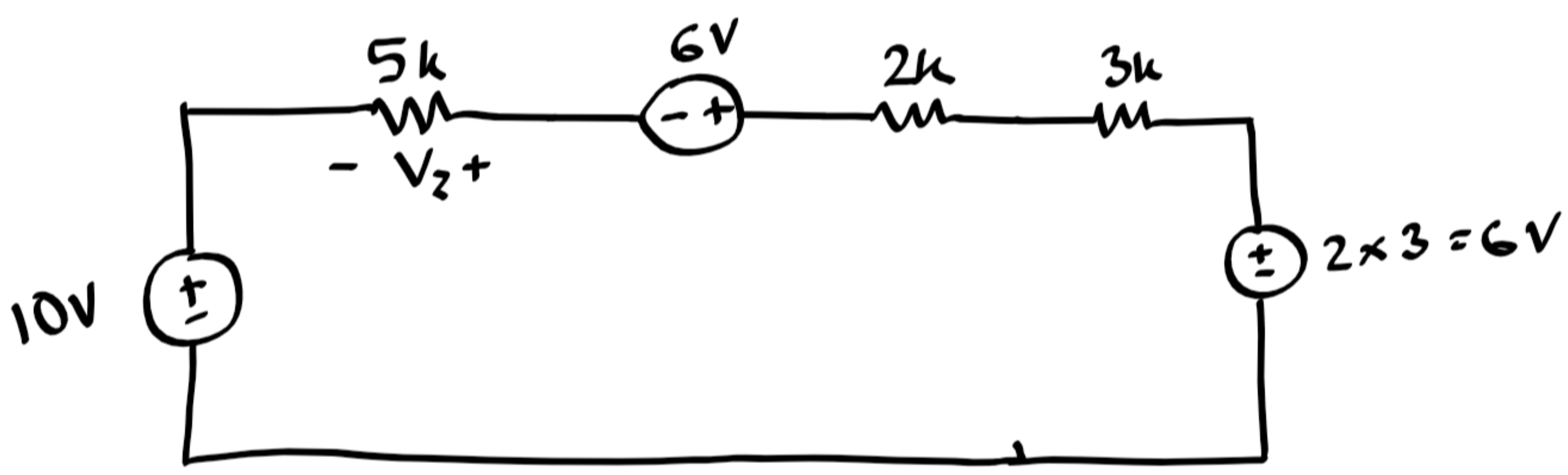
(supplied)

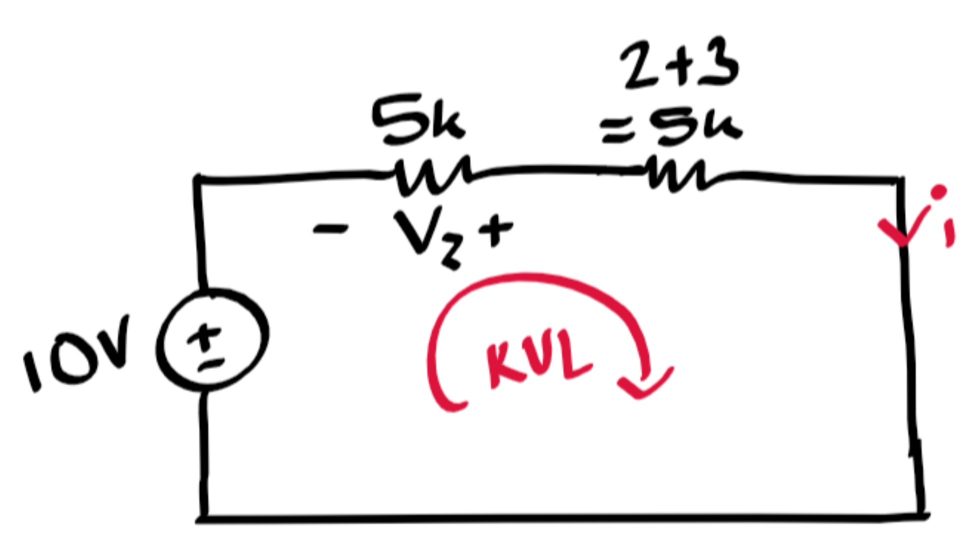
3)




KVL,

$-10 + 5i + 5i = 0$

$i = 1\ mA$

$\therefore V_z = -5i$

$= -5V$

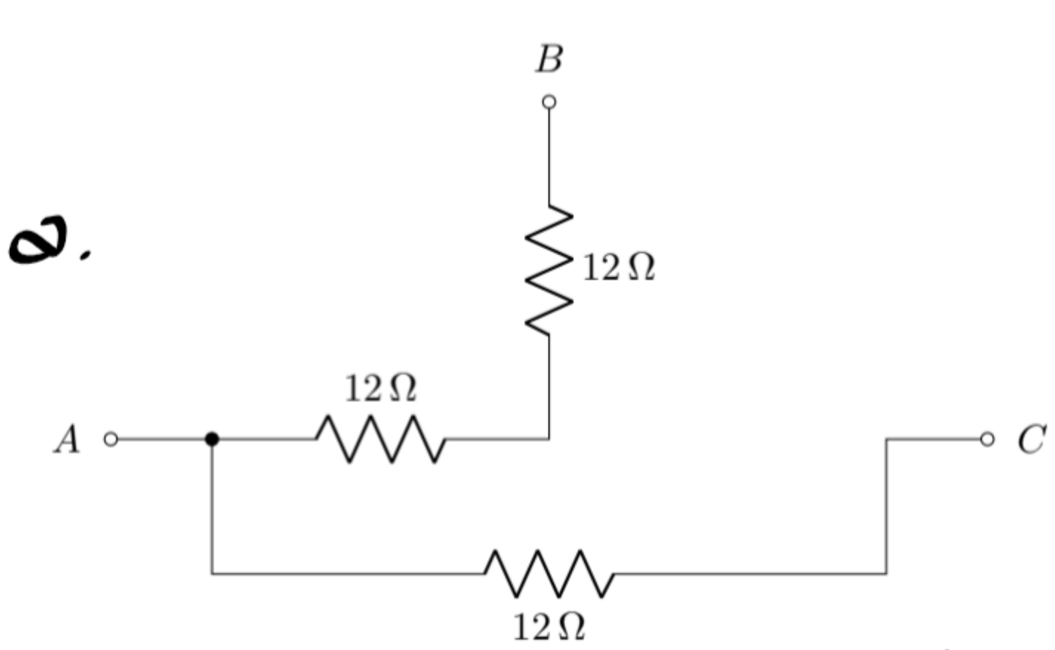
or

VDR,

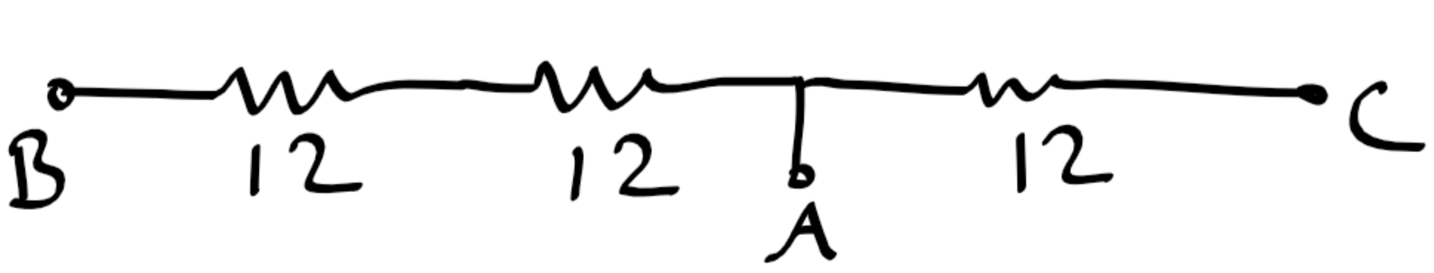
$V_z = -\frac{5}{5+5} \times 10$

$= -5V$

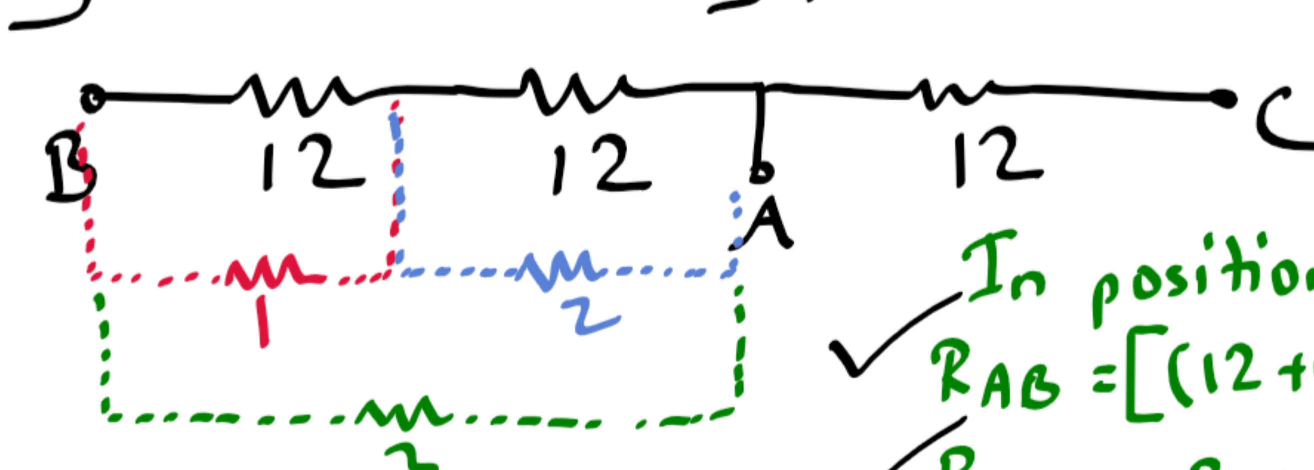
4) a.




For ease of visualization, straighten out the configuration-




Connecting the new resistor across nodes A-C would change the given $R_{AC} = 12\ \Omega$, so the resistor connected would have to be in any one of the following positions-




In position 3,

$R_{AB} = [(12+12) \| 12] = 8$

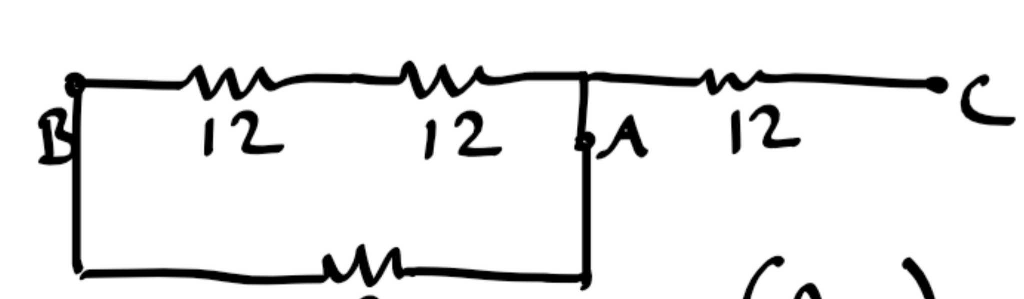
$R_{BC} = R_{AB} + 12 = 20$

In position 1,

$R_{AB} = (12 \| 12) + 12 = 18$

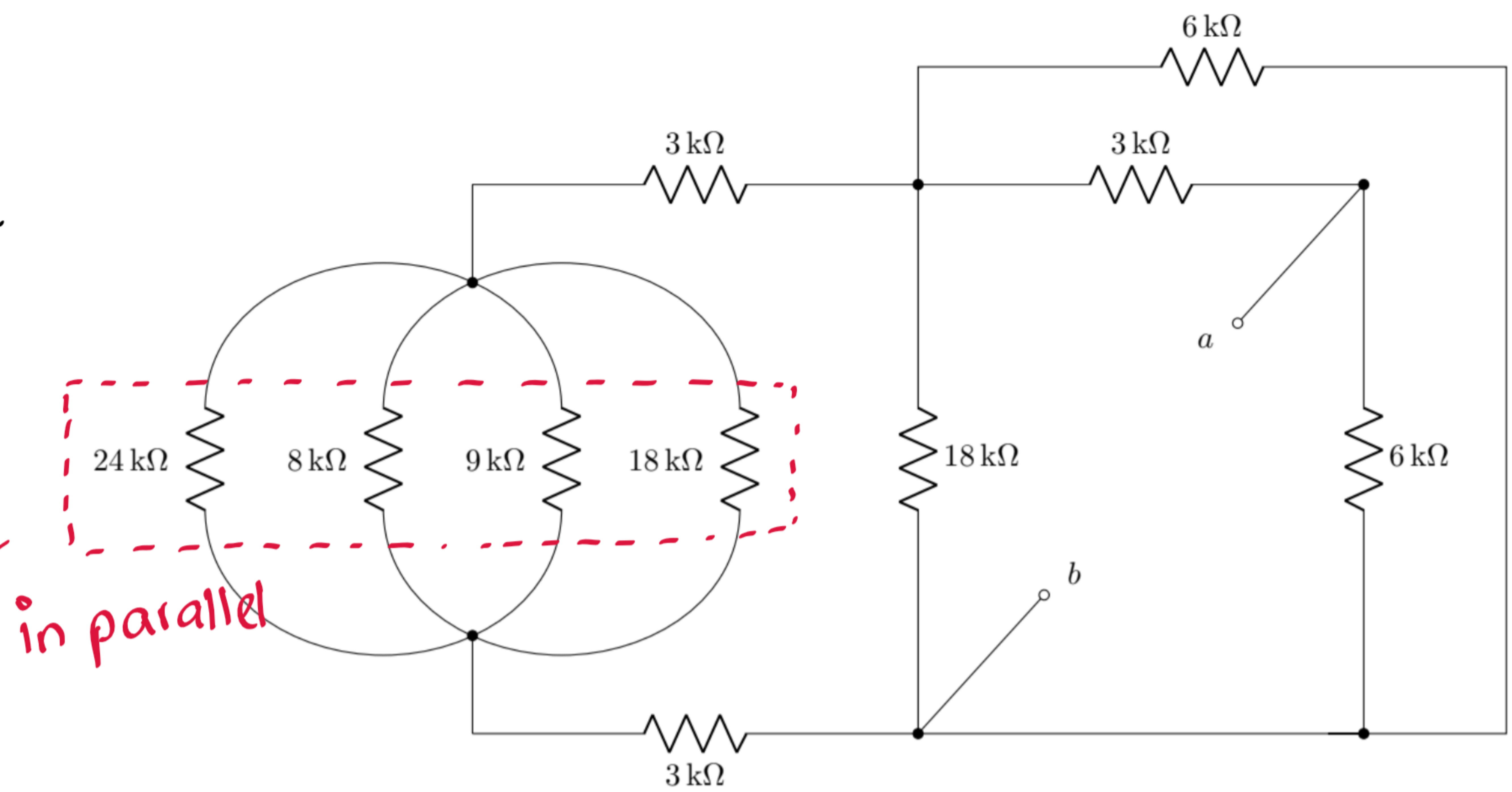
In position 2,

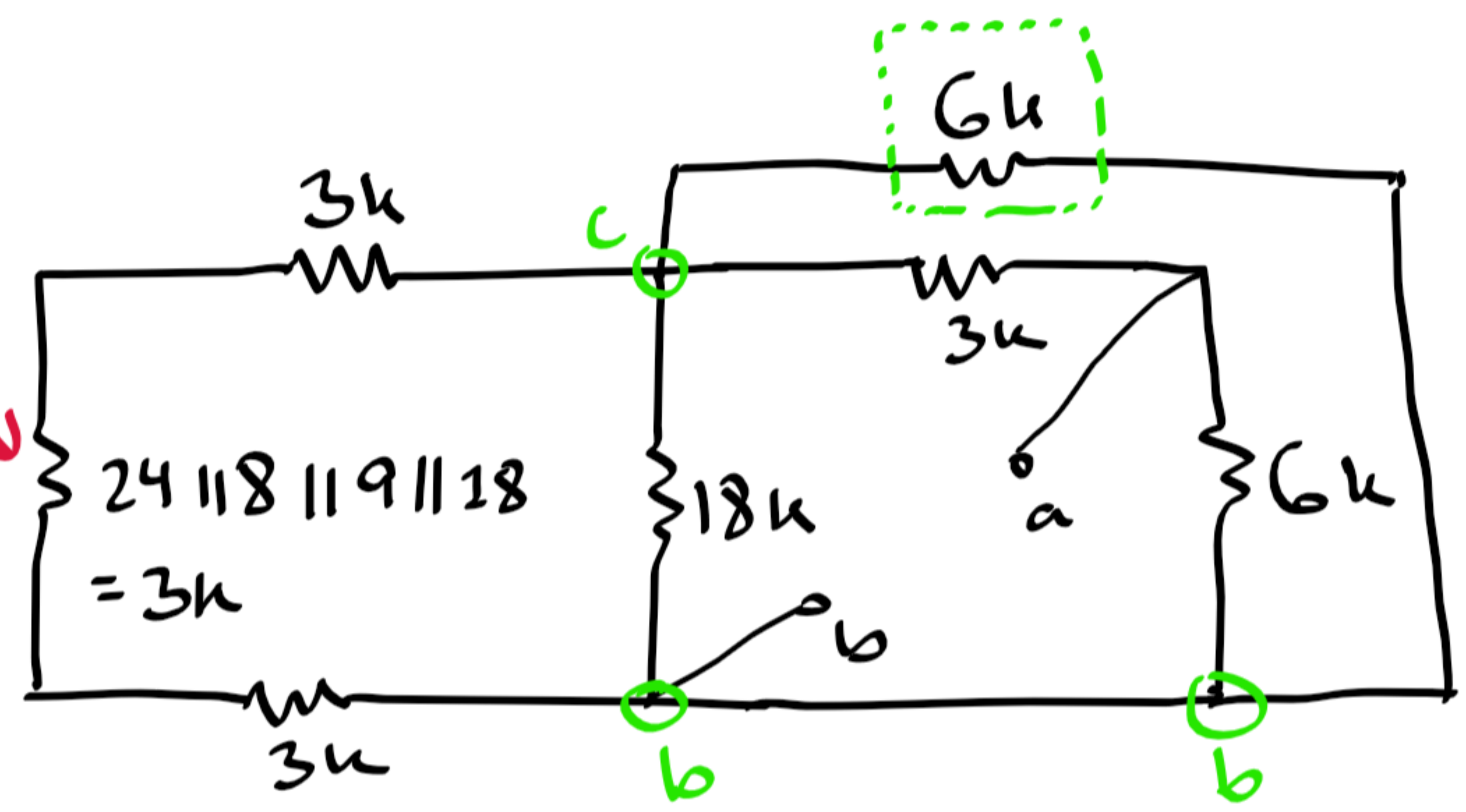
$R_{AB} = 12 + (12 \| 12) = 18$

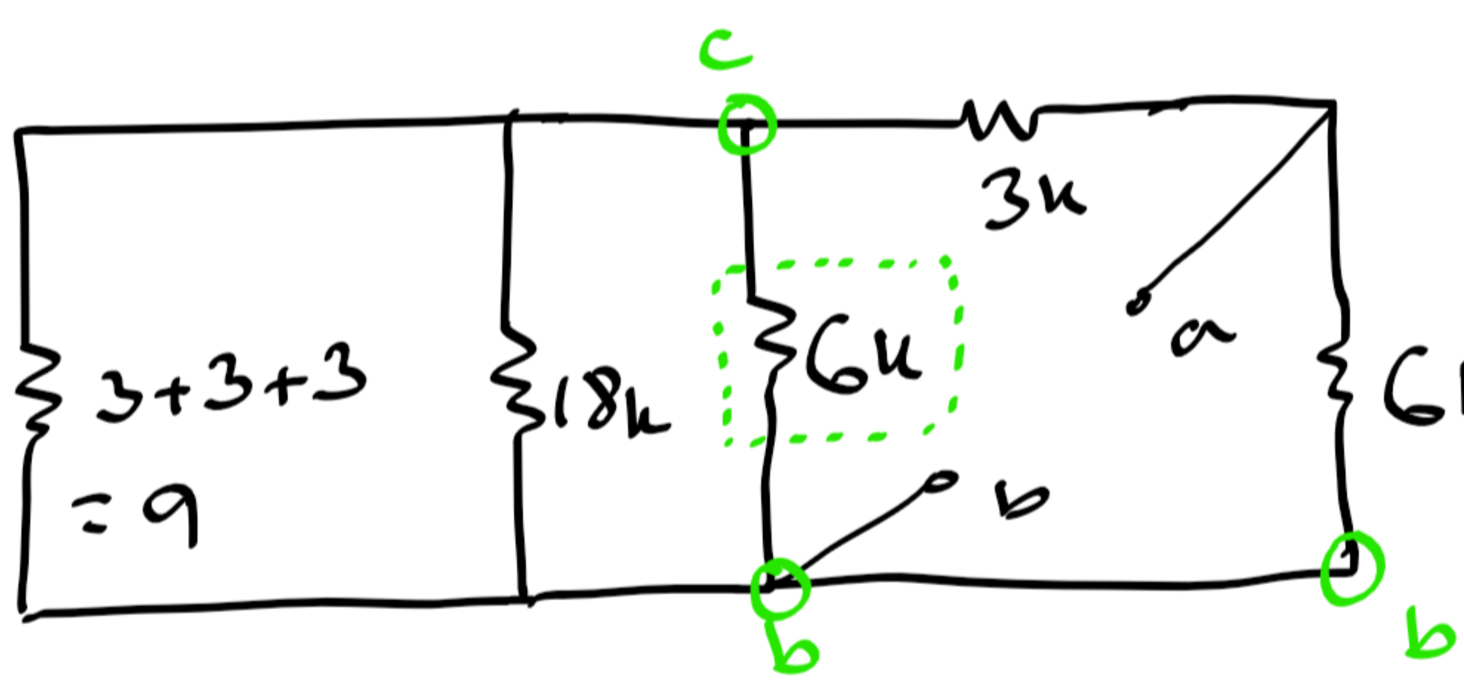


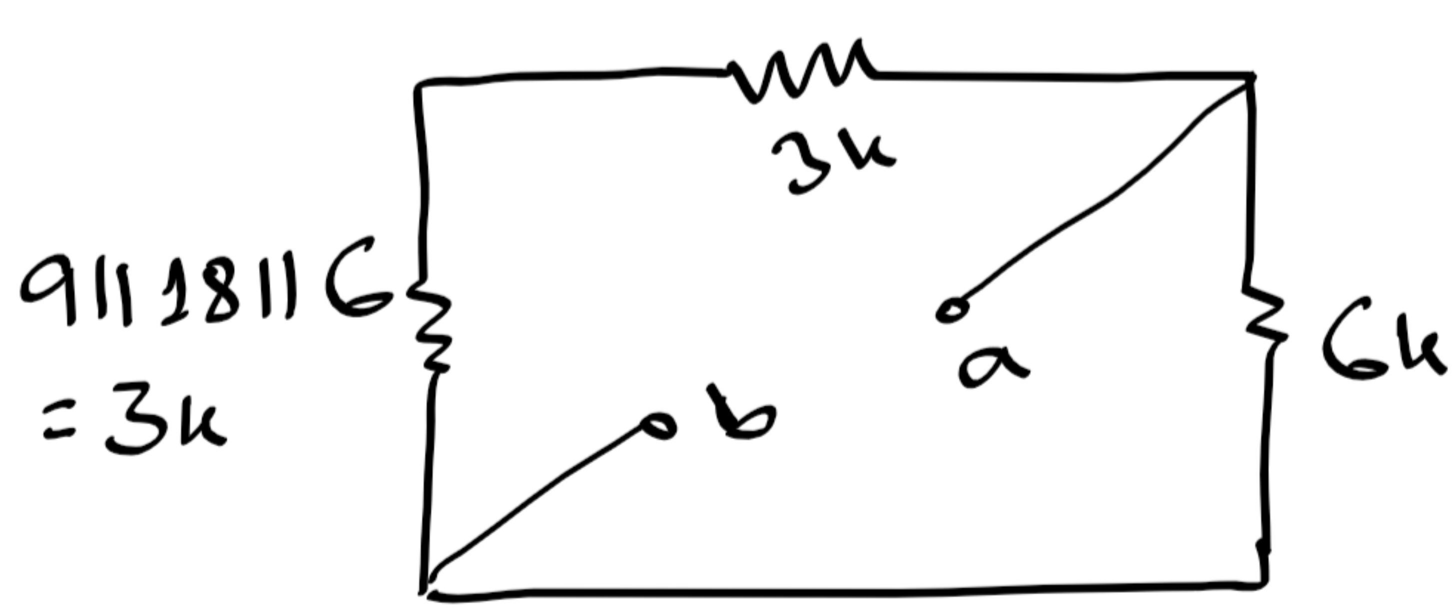

(Ans)

4) b.



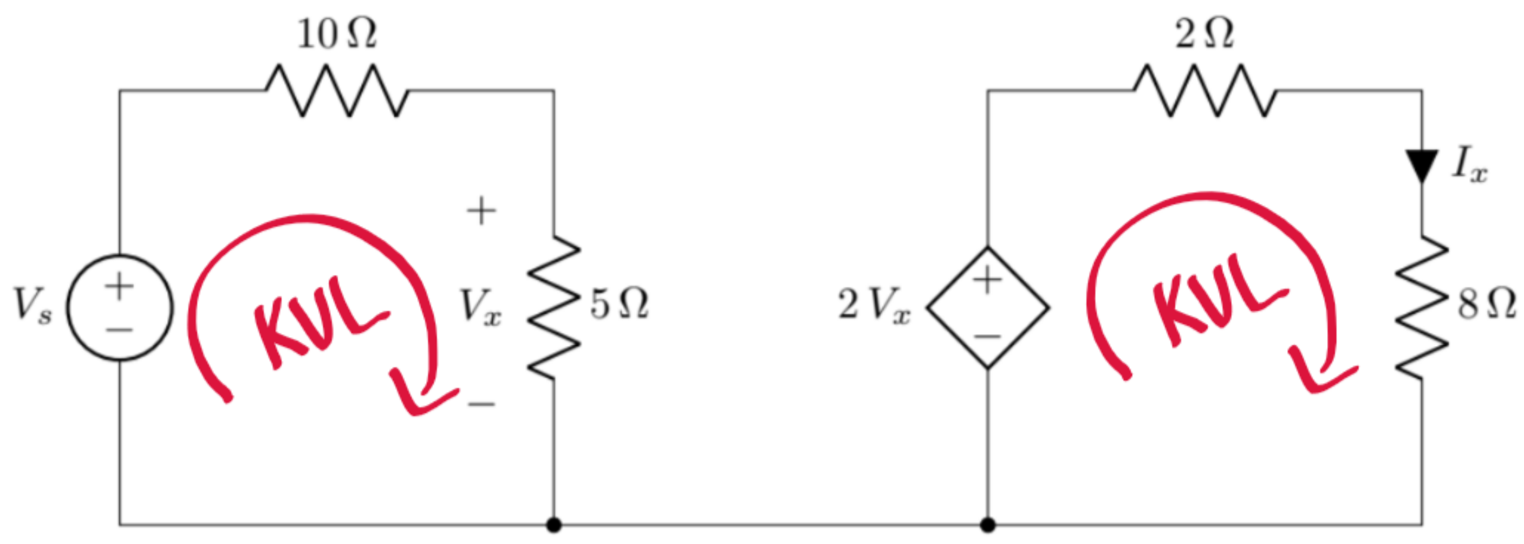




$$\therefore R_{ab} = (3+3) \parallel 6 = 3\,k\Omega$$

## Set B

1)




a. $P = 8\,W$

Using $P = I^2 R$,

$\Rightarrow 8 = I_x^2 \times 8$

$\boxed{\therefore I_x = 1\,A}$

c. Apply VDR in left mesh,

$\Rightarrow V_x = \dfrac{5}{5+10} \times V_s$

$\boxed{\therefore V_s = 15\,V}$

b. KVL in right mesh,

$-2V_x + 2I_x + 8I_x = 0$

$\therefore \boxed{V_x = 5V}$

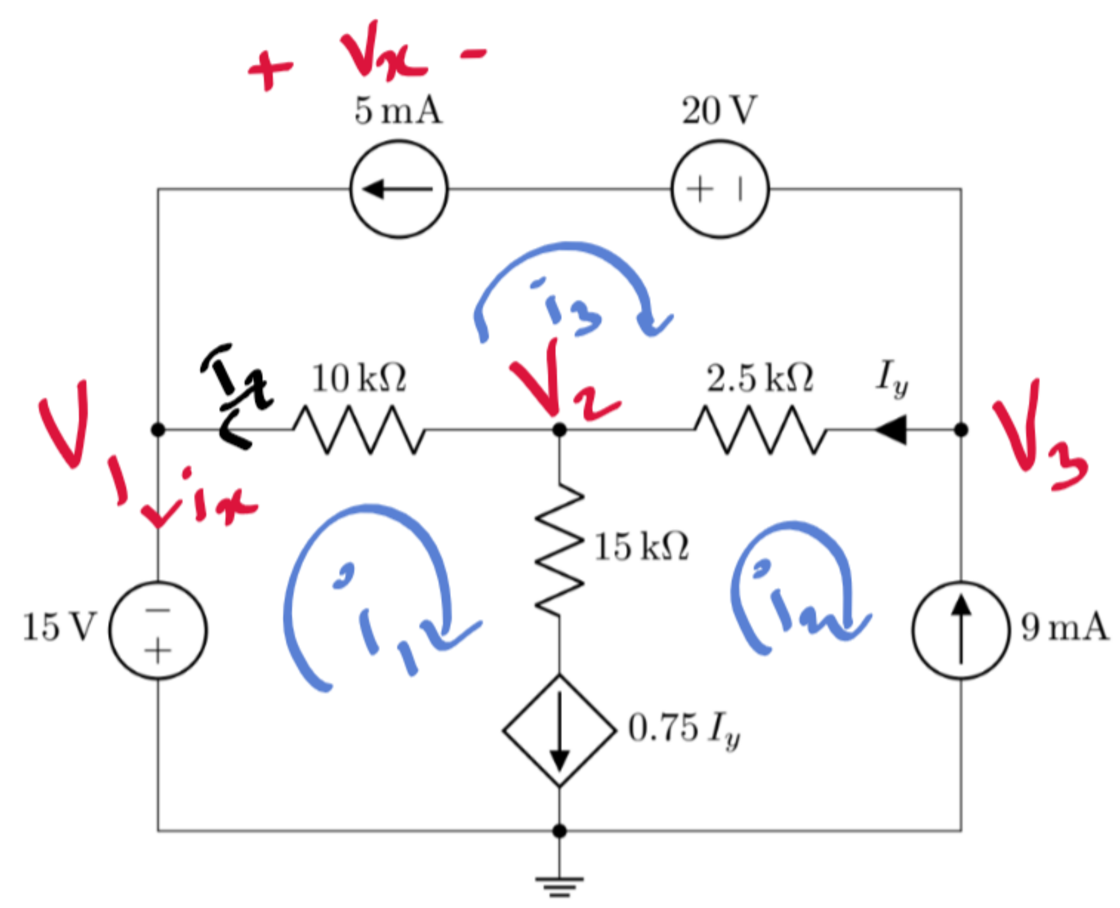
or use VDR,

$V_{8\Omega} = \dfrac{8}{2+8} \times 2V_x$

$\Rightarrow 8I_x = \dfrac{8}{10} \times 2V_x$

$\Rightarrow V_x = 5V$

2)




b. see part (a).

a. Method 1 : Nodal Analysis

$V_1$ : $V_1 = -15$ — (i)

$V_2$ : $\dfrac{V_2 - V_1}{10} + 0.75\,I_y + \dfrac{V_2 - V_3}{2.5} = 0$

$\Rightarrow -2V_1 + 10V_2 - 8V_3 = 15\,I_y$ — (ii) becomes $-V_1 + 2V_2 - V_3 = 0$

$V_3$ : $(-9) + \dfrac{V_3 - V_2}{2.5} + 5 = 0$

$\Rightarrow V_3 - V_2 = 10$ — (iii)

$I_y = \dfrac{V_3 - V_2}{2.5}$ — (iv)

solving,

$\boxed{V_1 = -15V,\ V_2 = -5V,\ V_3 = 5V}$

Method 2 : Mesh Analysis

(2) : $i_2 = -9$ — (i)

(3) : $i_3 = -5$ — (ii)

(There is no need to write an equation for mesh 1)

and KCL equations -

(3 and 2) $I_y = i_3 - i_2$ — (iii)

(1 and 2) $0.75\,I_y = i_1 - i_2$ — (iv) becomes $4i_1 - i_2 - 3i_3 = 0$

solve (iv) -

$\therefore i_1 = -6\,mA$
and $i_2 = -9\,mA$
$i_3 = -5\,mA$

i) $P_{5mA} = -5 \times V_x$

nodal: $= -5 \times (V_1 - V_3 - 20)$
$= 200\,mW$
(absorbed)

mesh: $= -5\left[-20 - 10(i_3 - i_1) - 2.5(i_3 - i_2)\right]$
$= 200\,mW$
(absorbed)

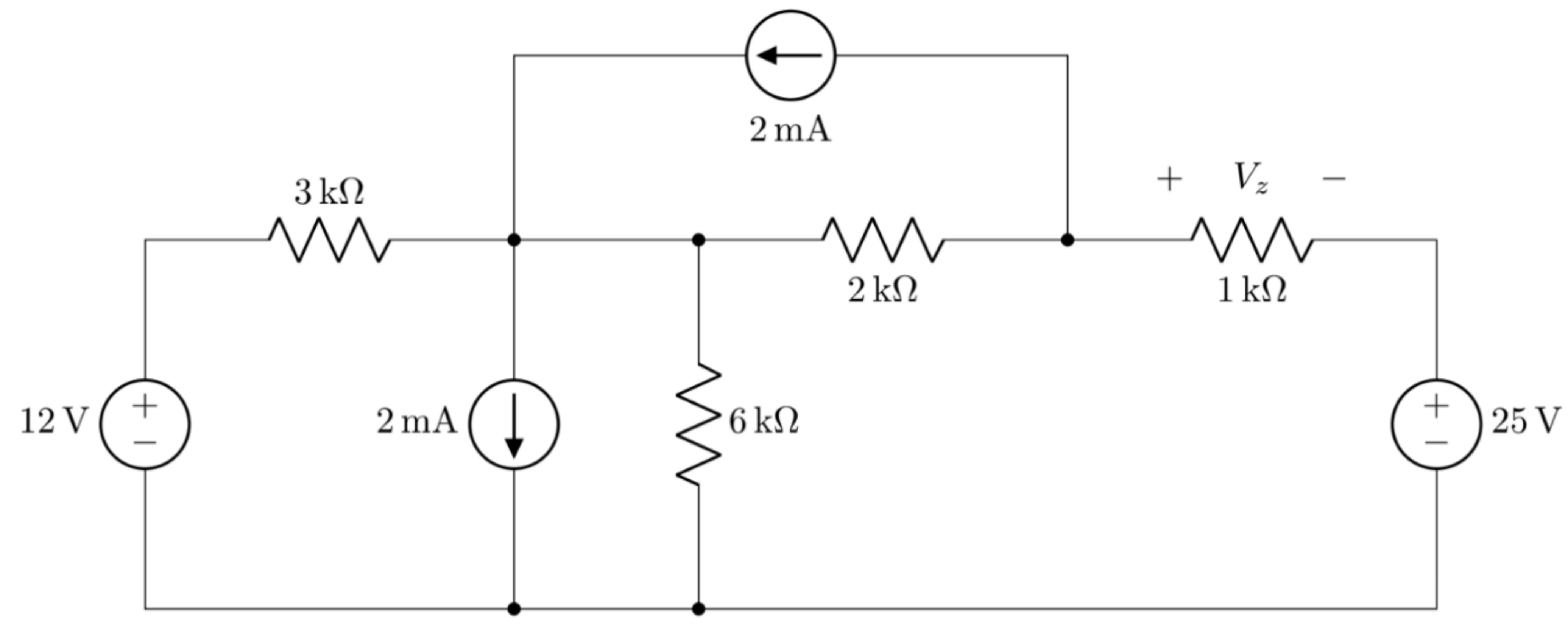
ii) $P_{15V} = -i_x \times 15$

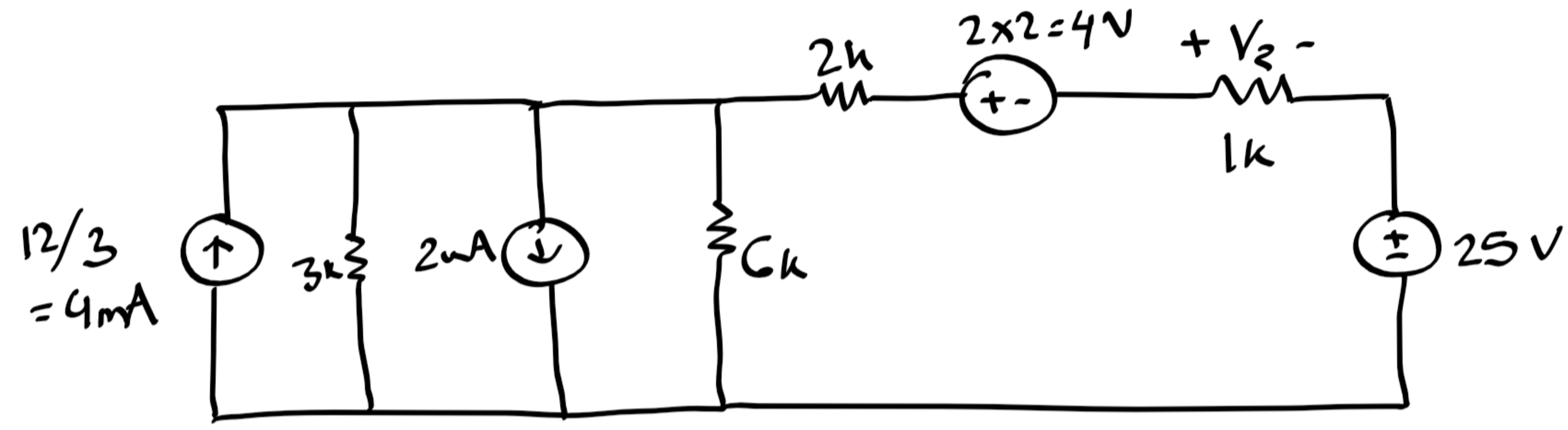
$I_z = \dfrac{V_2 - V_1}{10}$

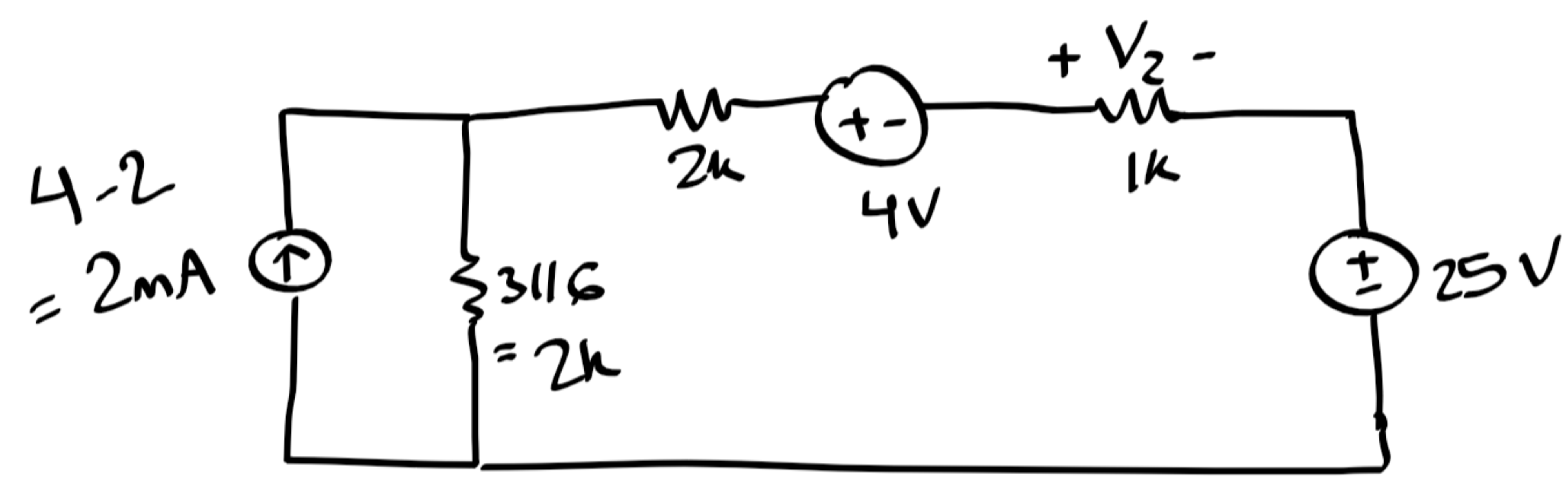
nodal: $= -(5 + I_z) \times 15$
$= -90\,mW$
(supplied)

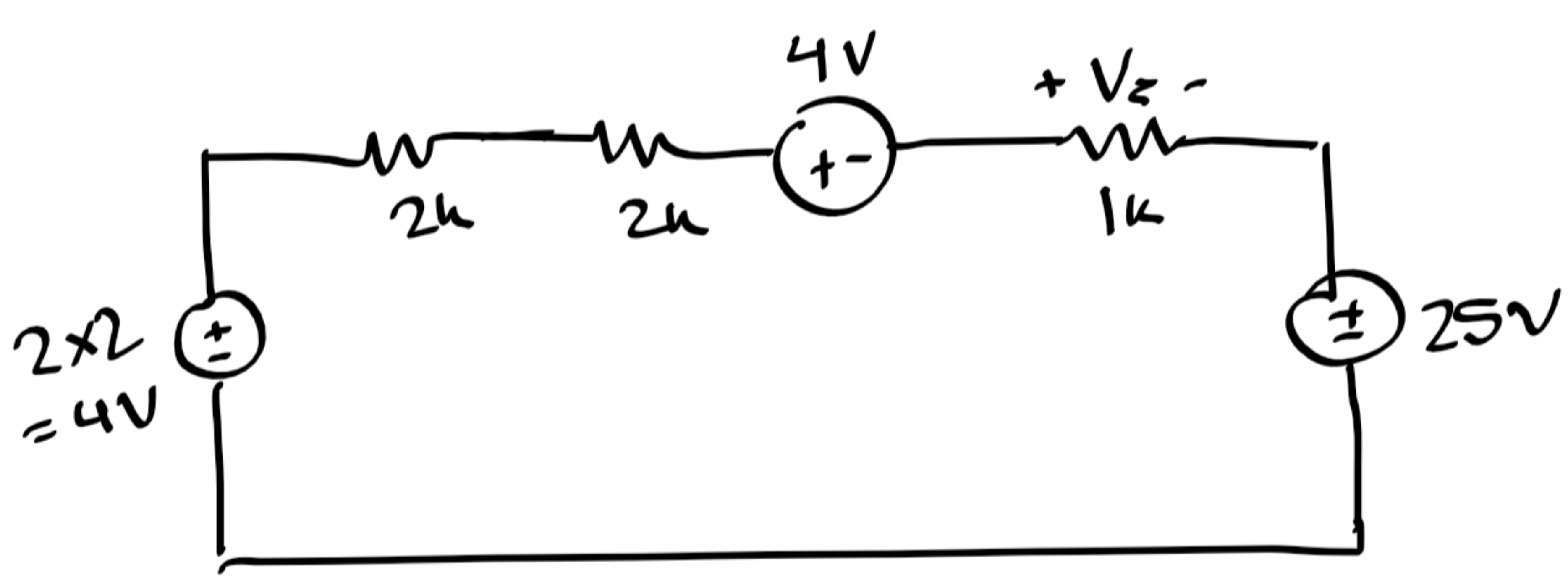
mesh: $= -(-i_1) \times 15$
$= -90\,mW$
(supplied)

3)

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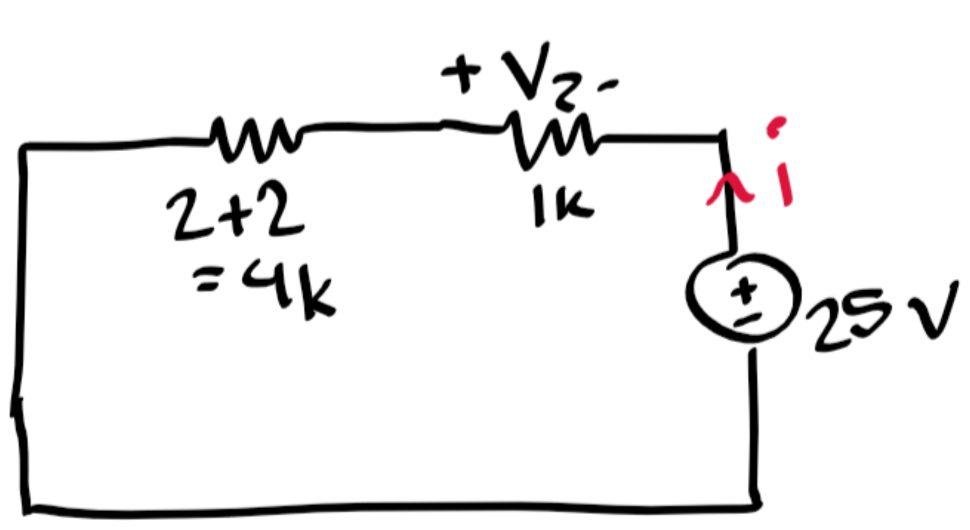



KVL,

$-4i - 1i + 25 = 0$

$i = 5\,mA$

$\therefore V_z = -1\,i$

$= -5V$

or

VDR,

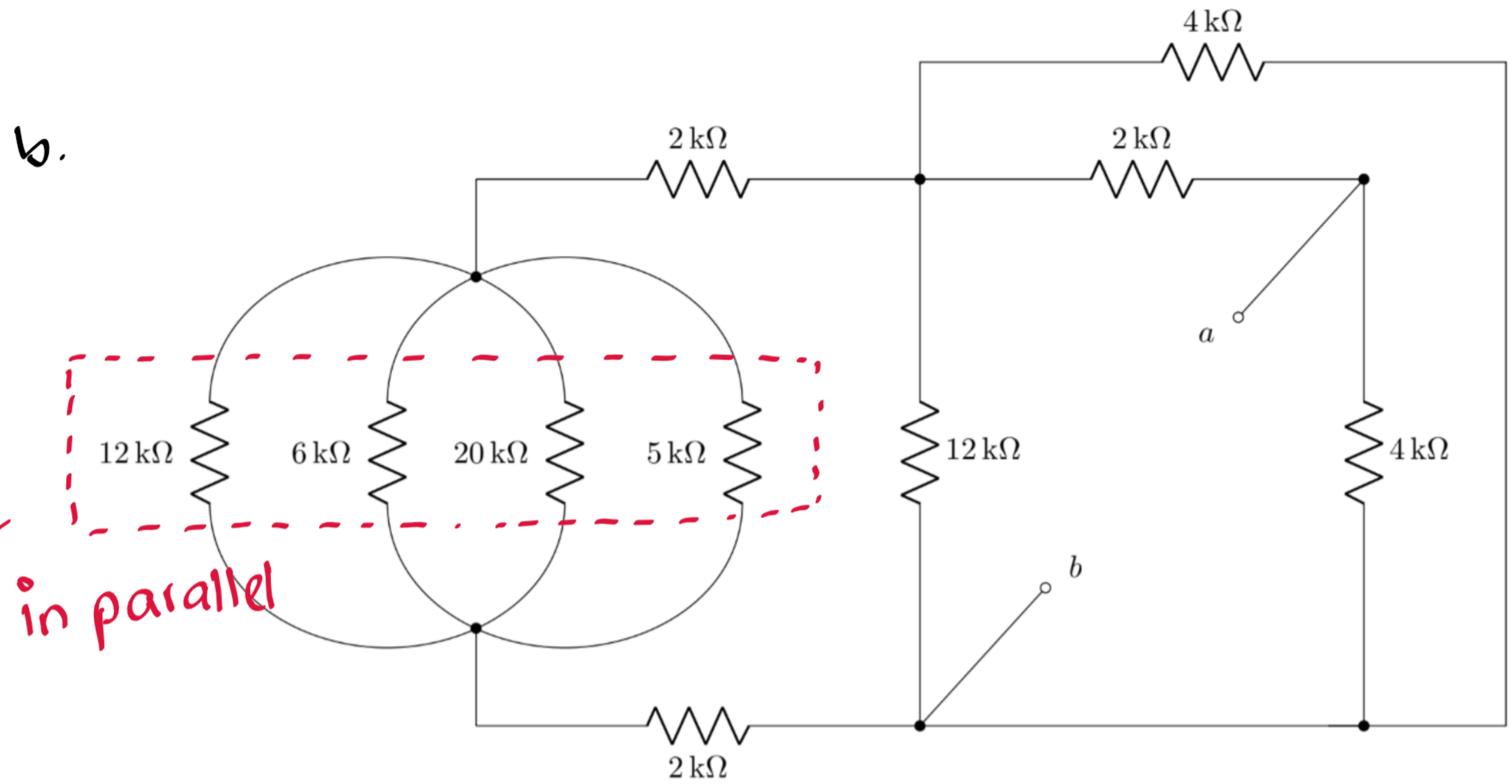
$$V_z = -\frac{1}{1+4} \times 25$$

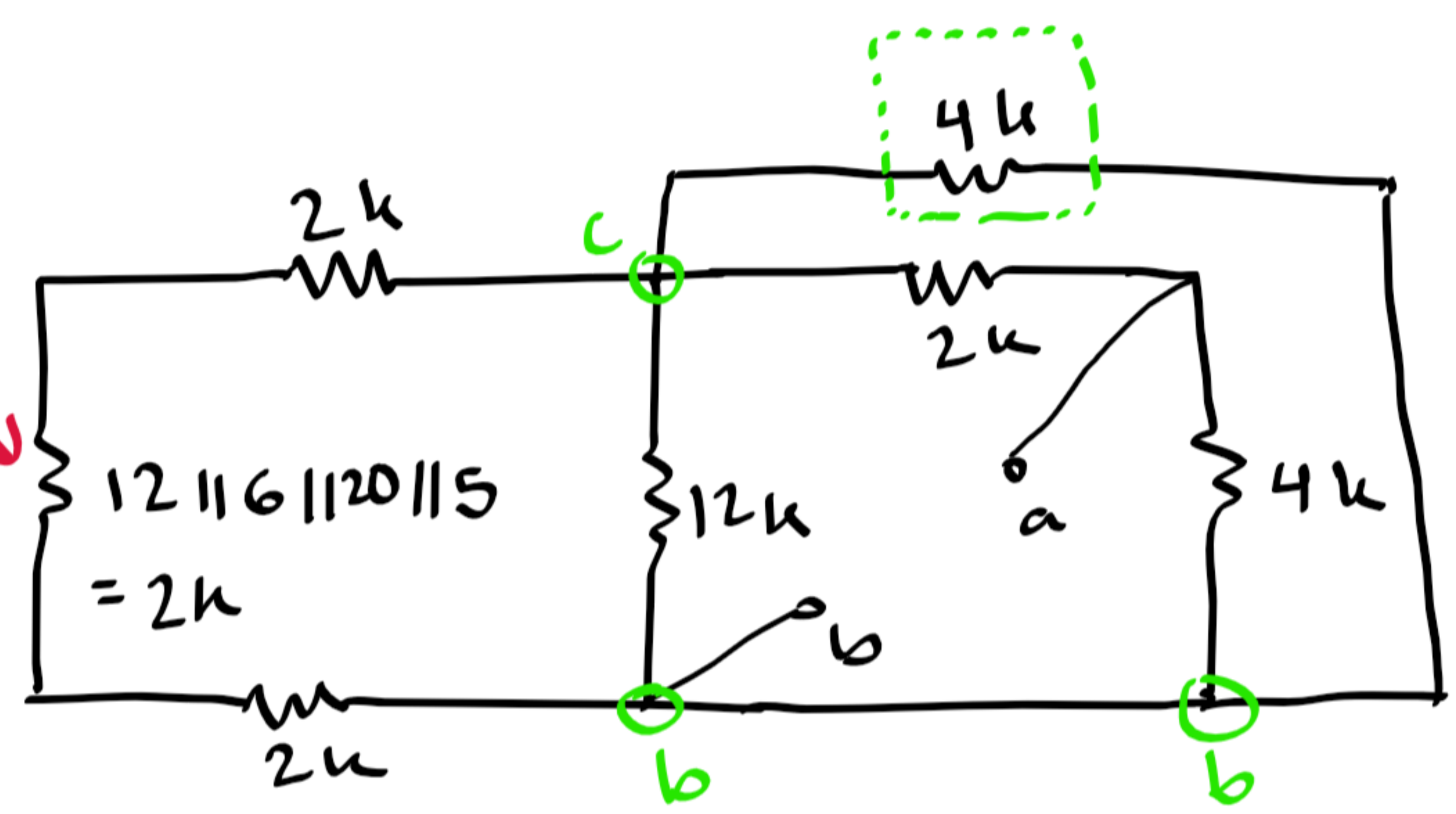
$$= -5V$$

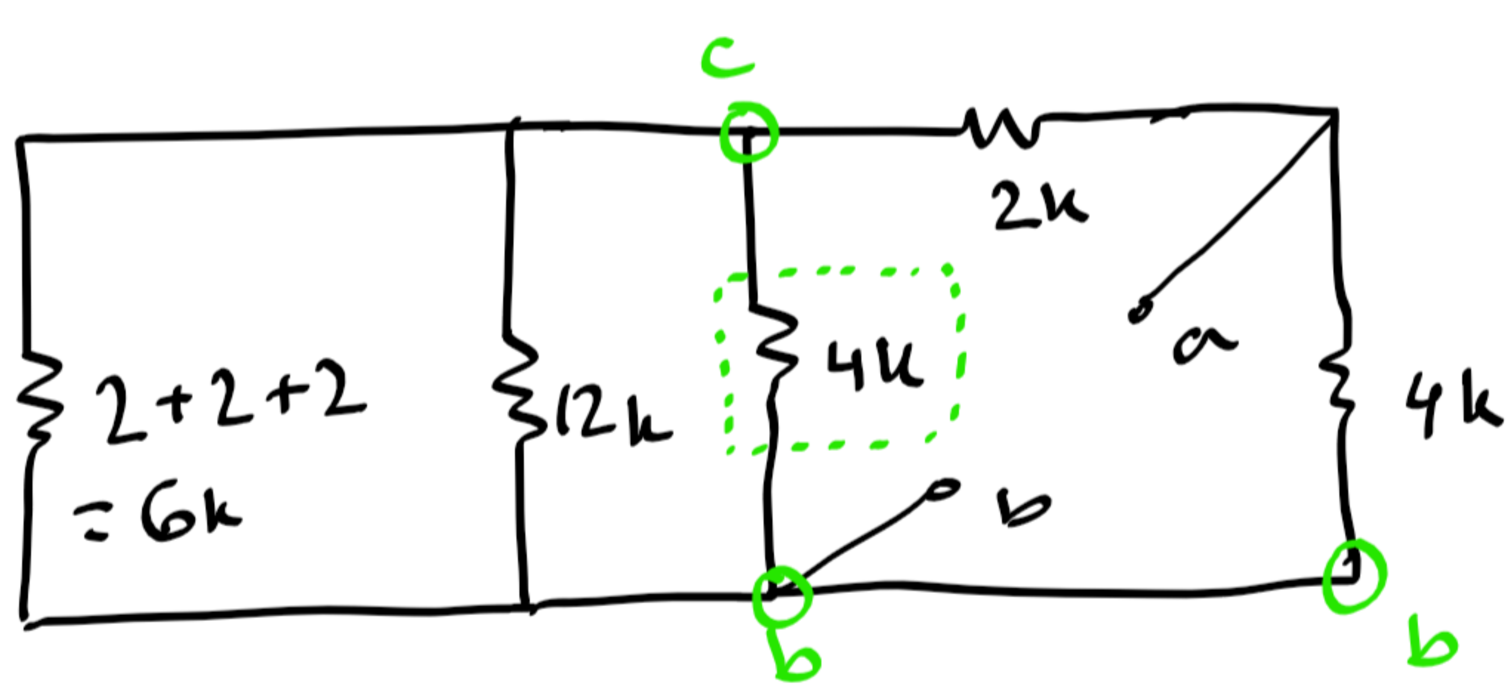
4) a. same as set A.

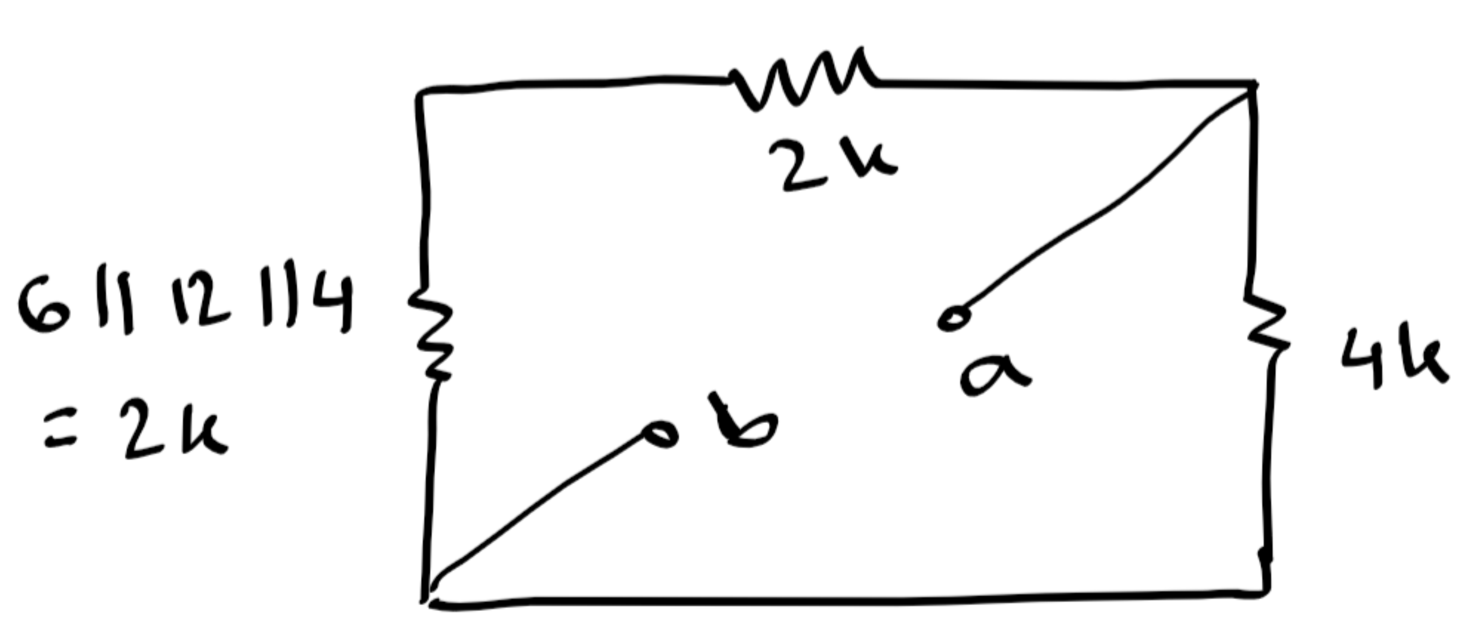
(nodes A and C have been interchanged)

4) b.




$\therefore R_{ab} = (2+2) \parallel 4 = 2\,k\Omega$